Revisiting the Two-Ray Propagation Modeling for Short-Range WiFi Overwater Links

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***Abstract*—This paper revisits the use of the two-ray propagation model for short-range WiFi links at 2.4 GHz over a freshwater lake. Experimental measurements are used to compare the simplified and complete forms of the model. The complete formulation, which incorporates the Fresnel reflection coefficient, complex relative permittivity, and polarization, achieves a Root Mean Squared Error (RMSE) of 1.44 dB and a Mean Absolute Error (MAE) of 1.07 dB after compensating for transmit power offsets. These errors are approximately 66% lower than those of the adjusted simplified version (RMSE 4.28 dB, MAE 2.85 dB). The results confirm that the complete formulation provides a more accurate basis for modeling short-range overwater propagation and IoT-oriented link design, while also refining and improving prior analyses of the same experimental dataset.**

***Index Terms*—Two-ray model, overwater propagation, WiFi.**

I. Introduction

Short-range overwater WiFi links are increasingly used in Internet-of-Things (IoT) monitoring, aquatic sensing, and autonomous platforms [1], [2]. Their behavior is strongly affected by specular reflections at the water surface, antenna geometry, and polarization, which directly influence communication range and reliability. The two-ray model is widely regarded as an effective predictor of these effects and is therefore the standard tool for overwater propagation modeling [3].

While the simplified form of the model [4] remains widely used in the literature [3], [5], it assumes ideal reflection for linear (vertical or horizontal) polarization and overlooks the dielectric properties of water. This is often a modeling choice rather than a misconception, but it can lead to noticeable accuracy differences at short ranges, an aspect often ignored in practical analyses. The complete formulation [6], which incorporates the permittivity-dependent Fresnel reflection coefficient, provides a more realistic representation of the interference pattern, even under tidal fading conditions [7].

This work revisits a 2.4 GHz dataset collected over a freshwater lake and quantifies the improvement achieved by the complete formulation after compensating transmit power offsets. The results validate the applicability of the full two-ray model in this environment and show that incorporating polarization and permittivity features significantly enhances modeling accuracy, even with censored RSSI measurements.

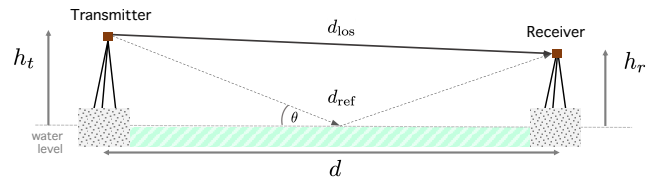

Fig. 1: The two-ray model geometry showing the direct line-of-sight path (d_{los}) and the path reflected on the water surface (d_{ref}).

II. Background on the Two-Ray Model

For completeness, we include the expressions of the two-ray model in its simplified and complete forms. The model describes the received power resulting from a direct path and a path reflected on the water surface. Fig. 1 shows the geometry of the Shore-to-Shore (S2S) link, where the Line-of-Sight (LoS) and reflected paths have lengths d_{los} and d_{ref}. The reflection occurs at an incidence angle θ. The transmitter (TX) and receiver (RX) heights above the water surface are h_t and h_r, and d is their horizontal separation. The path length difference produces a phase shift $\Delta_\phi = 2\pi(d_{\mathrm{ref}} - d_{\mathrm{los}})/\lambda$, with λ being the wavelength.

Two-ray model (complete) [6]. The received power is given by

$$P_r = P_t G_t G_r \left(\frac{\lambda}{4\pi}\right)^2 \left| \frac{1}{d_{\mathrm{los}}} + \Gamma \, \frac{e^{-j\Delta_\phi}}{d_{\mathrm{ref}}} \right|^2 \tag{1}$$

where P_t is the transmit power and G_t, G_r are the TX and RX antenna gains. For vertical polarization, the Fresnel reflection coefficient is written as $\Gamma = (\sin\theta - Z)/(\sin\theta + Z)$, with $Z = \sqrt{\varepsilon_c - \cos^2\theta}/\varepsilon_c$. The water medium is characterized by the complex relative permittivity $\varepsilon_c = \varepsilon_r - j\,\sigma/(\omega\varepsilon_0)$, where ε_r is the relative permittivity, σ the conductivity, ε_0 the vacuum permittivity, and c is the speed of light.

Two-ray model (simplified) [4]. This expression assumes that the magnitude of Γ is unitary, reducing Eq. (1) to

$$P_r = P_t G_t G_r \left(\frac{\lambda}{4\pi}\right)^2 \left(2\sin\left(\frac{2\pi h_t h_r}{\lambda d}\right)\right)^2 \tag{2}$$

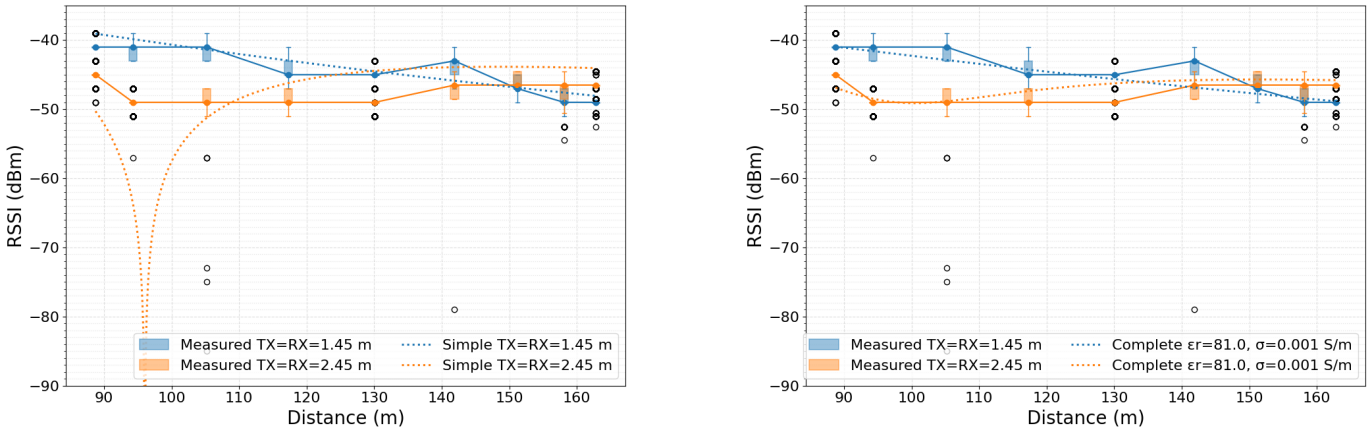

Fig. 2: Comparison between the experimental data and the two-ray model: (a) simplified and (b) complete versions.

TABLE I: RMSE and MAE for the simplified and complete two-ray models using a single optimal offset ΔP_t per model, for each antenna height and for the combined dataset.

Model	Height (m)	ΔP_t (dBm)	RMSE (dB)	MAE (dB)
Simple	1.45	4.57	1.52	1.26
	2.45		5.85	4.44
	Combined		4.28	2.85
Complete	1.45	4.89	1.52	1.03
	2.45		1.35	1.10
	Combined		1.43	1.07

III. Evaluation

A. Measurement Dataset. We use the overwater measurements reported in [3] where transmissions were made on IEEE 802.11b channel 1 at $f = 2412$ MHz with $P_t = 20$ dBm and vertically oriented omnidirectional antennas ($G_t = G_r = 5$ dBi). The campaign includes nine S2S links with distances between about 88 and 163 m, measured at two antenna heights (1.45 and 2.45 m). Note only the parameters relevant to the modeling are summarized here; full details are in [3].

B. Modeling Assumptions. We assume a complex relative permittivity with $\varepsilon_r = 81$ and $\sigma = 0.001$ S/m for freshwater [4]. Cable and hardware losses are also neglected, so the applied TX power offset represents a standard empirical calibration at a reference distance d_0. Channel is also assumed symmetric, and antennas with isotropic radiation patterns.

C. Results. Table I summarizes the accuracy obtained after applying the optimal transmit-power offset ΔP_t to each model. For the combined dataset, the simplified model yields an RMSE of 4.28 dB and an MAE of 2.85 dB, whereas the complete formulation reduces these values to 1.43 dB and 1.07 dB, improvements of about 66% and 63%, respectively.

Fig. 2 illustrates these differences across distance. The simplified model (*left*) reproduces the general trend but fails to capture the interference pattern and exhibits larger deviations, particularly for the 2.45 m height. The complete formulation (*right*) provides a consistently closer match to the measurements across all distances and both antenna heights, reflecting the quantitative improvements observed in the table.

IV. Conclusion

This work revisited short-range overwater WiFi propagation and showed that the complete two-ray formulation, which incorporates polarization and the Fresnel reflection coefficient, fits measured RSSI far better than the simplified expression. After a standard transmit-power correction, it reduced RMSE and MAE by about 65–75% across heights and in the combined dataset. Although the study covers a single scenario, future work will extend the analysis to other conditions to identify when the gain of the full model is more pronounced. Since both forms have similar computational cost and the simplified model offers no benefit beyond fewer parameters, the complete formulation stands as the more suitable choice for short-range overwater links and IoT-oriented design.

Acknowledgments

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